

WILLIAM BROWN

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SUMMARY

With 9+ years of experience in software engineering across diverse sectors such as internet services, legal/government, healthcare, finance, mobile technology, education, and transportation. Specializes in AI and ML, with a focus on developing conversational applications, speech recognition, and natural language processing. Skilled in applying machine learning techniques to tackle complex problems in data analysis.

SKILLS

- ✓ **Artificial Intelligence (AI) and Machine Learning (ML):** Expertise in leveraging AI and ML for building advanced applications, including conversational interfaces and speech recognition systems. Skilled in Generative AI applications and techniques.
- ✓ **Natural Language Processing (NLP):** Skilled in implementing NLP techniques to enhance machine understanding of human language in various applications.
- ✓ **Data Analysis:** Experienced in using machine learning algorithms to solve complex security challenges and perform sophisticated data analysis.
- ✓ **Web Development:** Proficient in Python programming and full stack development, encompassing both front-end and back-end technologies, frameworks, and tools across various industries.
- ✓ **Data Engineering:** Adept in designing, constructing, installing, testing, and maintaining highly scalable data management systems. Experienced in developing algorithms to transform data into actionable insights.
- ✓ **Industry Knowledge:** Strong understanding of multiple sectors including internet services, legal/government, healthcare, finance, mobile technology, education, and transportation, enabling tailored software solutions.
- ✓ **Technical Problem Solving:** Capable of diagnosing and resolving complex technical issues with innovative solutions.

EXPERIENCE

AI/ML Engineer

Cohere

02/2022 – Present

- Enhanced machine learning pipelines for robust real-time defect detection in production lines, significantly improving manufacturing efficiency and accuracy.
- Led a team in building generative AI models for creating realistic training data, enhancing the effectiveness of computer vision systems.
- Collaborated on interdisciplinary projects to integrate AI with existing technologies, improving predictive maintenance capabilities in manufacturing.
- Responsible for optimizing deep learning algorithms for faster processing and better accuracy, utilizing state-of-the-art computer vision techniques.
- Initiated and led the development of an automated visual inspection system, integrating deep learning models that outperformed traditional methods.
- Deployed scalable AI solutions across multiple production environments, ensuring seamless operation and maintenance of machine learning models.

- Reduced error rates in automated systems by 20% through continuous refinement of machine learning algorithms, focusing on quality control and defect detection.

Senior Natural Language Processing Engineer

InVideo

06/2020 – 12/2021

- Implemented NLP techniques to enhance machine understanding of human language, improving system interactions within an educational tech platform.
- Led the development of a large language model (LLM) to refine content personalization algorithms, achieving a significant increase in user engagement.
- Conducted advanced research in NLP to develop models that better interpret contextual nuances in text, enhancing natural language understanding.
- Optimized NLP pipelines for higher accuracy in sentiment analysis, contributing to refined customer feedback interpretations in the service sector.
- Developed and deployed chatbot solutions using NLP to automate and enhance customer service interactions, leading to a 40% increase in customer satisfaction.
- Spearheaded a team that integrated LLM capabilities into existing products, significantly reducing the time required for content generation.
- Employed feature engineering and NLP techniques for effective topic modeling on customer reviews, providing actionable insights for product development.

Data Scientist

Salesforce

07/2017 – 05/2020

- Analyzed large-scale volumes of real-time traffic data, processing 50M+ transactions during a major online shopping event, enhancing data-driven decision-making.
- Designed and implemented data science dashboards using Dash Plotly and Streamlit Python, facilitating real-time data visualization and insights.
- Conducted in-depth data analysis using complex SQL and Python, deriving actionable insights from vast datasets to inform strategic marketing campaigns.
- Developed machine learning models for predictive analytics, significantly improving forecast accuracy and operational efficiency in logistics.
- Created robust algorithms for data transformation, turning raw data into valuable insights that drove product development and customer engagement strategies.
- Integrated AI-driven analytics into business operations, enhancing the accuracy and speed of reporting and supporting data-driven decision-making at all levels.

AI Technical Support Specialist

UnitedHealth

11/2016 – 08/2017

- Provided technical support for AI-driven applications, resolving 80% of reported issues, significantly improving system reliability and user satisfaction.
- Assisted in the deployment and maintenance of AI systems, leading to a 40% reduction in system downtime and a 50% increase in user load handling capacity.
- Conducted performance testing and optimization of AI models, enhancing scalability and efficiency, and ensuring robust application performance under high traffic.
- Trained end-users on the effective use of AI tools, enhancing their ability to leverage technology for better productivity and insights.
- Monitored and visualized model operational metrics using Prometheus and Grafana, ensuring optimal performance and quick troubleshooting of AI applications.

- Collaborated with product development teams to relay user feedback and improve AI model accuracy and interface usability.
- Supported the integration of AI technologies into existing IT infrastructures, providing expert advice and practical solutions for seamless transitions.

Backend Engineer

UnitedHealth

09/2015 – 11/2016

- Developed robust web applications using Flask and Django, enhancing functionality and user experience for e-commerce platforms.
- Optimized backend operations using Django, significantly reducing server response times and improving scalability for high traffic applications.
- Utilized advanced features of Flask to develop lightweight microservices, enhancing modular architecture and easing the integration of third-party services.
- Deployed and maintained code using Docker and Azure DevOps CI/CD pipelines, ensuring continuous integration and delivery of backend applications.
- Designed database schemas and implemented ORM with Django, efficiently handling database operations and reducing development time.
- Collaborated with frontend teams to ensure seamless integration of APIs, providing robust backend support for interactive web applications.

EDUCATION

Stevens Institute of Technology, Hoboken, NJ

Bachelor of Science in Computer Science, 2015